

Linear systems and elimination

Sharp growth for cyclic tridiagonal matrices

Difficulty: challenging **Importance:** interesting to specialist

Status: open in the historical source; no resolution found in the bounded later search below.

Problem statement

For each $n \geq 4$, let C_n consist of the nonsingular matrices $A \in \mathbb{C}^{n \times n}$ whose only permitted nonzero positions satisfy

$$|i - j| \leq 1 \quad \text{or} \quad (i, j) \in \{(1, n), (n, 1)\},$$

with $a_{1n} \neq 0$ and $a_{n1} \neq 0$. This is the cyclic tridiagonal pattern called *quasi-tridiagonal* in the source. No symmetry or diagonal dominance is assumed.

Apply Gaussian elimination with partial pivoting (GEPP) in exact arithmetic. At each stage select a largest-modulus entry in the active first column, interchange its row with the active first row, and form the trailing Schur complement. For an admissible path π , let $S_1 = A, \dots, S_n$ denote the active matrices. Define

$$\rho(A, \pi) = \frac{\max_{1 \leq k \leq n} \|S_k\|_{\max}}{\|A\|_{\max}}, \quad \|M\|_{\max} = \max_{i,j} |m_{ij}|,$$

$$c_n = \sup_{A \in C_n} \sup_{\pi \text{ permitted by GEPP}} \rho(A, \pi).$$

Determine c_n for every $n \geq 4$, with matching upper bounds and examples approaching each supremum. All admissible tie choices are included. The given ordering is fixed before the stated GEPP run; preliminary row or column reorderings are not permitted. All sizes constitute one extremal problem.

Ordinary tridiagonal inputs have growth at most two. This question isolates the effect of the two corner entries on partial-pivoting stability. Unlike the fixed-bandwidth problem IE-13, the corner offsets grow with n .

Reference

N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, second edition, SIAM (2002), Problem 9.15(b), p. 193; Theorem 9.11, p. 173, supplies the ordinary tridiagonal comparison.

Status check

The book's sparsity pattern was checked directly. Its wording omits the field; this entry adopts $\mathbb{C}$ from the adjacent Theorem 9.11. Searches for *quasi-tridiagonal growth factor*, *quasi-tridiagonal partial pivoting*, and *Gaussian elimination cyclic tridiagonal growth* found no solution of this exact extremal question. A [2017 cyclic-reduction paper](#) uses a different matrix pattern and algorithm. Shah–Urschel's [August 31, 2026 revision](#), Theorems 2.2–2.3, does not determine this extremum. No recent explicit reaffirmation of the question's openness was found; absence of a located resolution is the limit of this check.